EVERY CLOSED SUBSET CARRIES A UNIQUE REDUCED INDUCED CLOSED SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2410-2453, following the statement at 2402-2408.

Editorial note. The 'it is clear' sheaf-of-ideals sentence (~2419-2420) is a locality reading of the pointwise vanishing condition, reproduced in the notes.

Allowed standard prerequisites (editorial).

- Radical ideals and Zariski topology. Algebra, Lemma algebra-lemma-Zariski-topology (cross-chapter, cited at ~2427, ~2450). Identifies I(U) with the radical ideal and recovers it from V(I).
- Affine reduced criterion. lemma-affine-reduced (same chapter, cited at ~2439, ~2447). Spec of a reduced ring is reduced and conversely.
- Closed subspace associated to a sheaf of ideals. definition-closed-subspace (same chapter, statement at 606, construction consumed at ~2435-2436). Defines Z from the quasi-coherent sheaf of ideals I.

Source reference. Original reference 00E0: focused-seeds/geometry/S9-schemes-reduced-closed-subscheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01J2: sources/stacks-schemes/pinned-original-source.tex, source lines 2389–2389. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. Let $T \subset X$ be a closed subset. There exists a unique closed subscheme $Z \subset X$ with the following properties: (a) the underlying topological space of Z is equal to T, and (b) Z is reduced.*

Proof. Let $\mathcal{I} \subset \mathcal{O}_X$ be the sub presheaf defined by the rule

$$\mathcal{I}(U) = \{f \in \mathcal{O}_X(U) \mid f(t) = 0 \text{ for all } t \in T \cap U\}$$

Here we use $f(t)$ to indicate the image of f in the residue field $\kappa(t)$ of X at t. Because of the local nature of the condition it is clear that $\mathcal{I}$ is a sheaf of ideals. Moreover, let $U = \mathrm{Spec}(R)$ be an affine open. We may write $T \cap U = V(I)$ for a unique radical ideal $I \subset R$. Given a prime $\mathfrak{p} \in V(I)$ corresponding to $t \in T \cap U$ and an element $f \in R$ we have $f(t) = 0 \Leftrightarrow f \in \mathfrak{p}$. Hence $\mathcal{I}(U) = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p} = I$ by Algebra, Lemma 00E0. Moreover, for any standard open $D(g) \subset \mathrm{Spec}(R) = U$ we

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have $\mathcal{I}(D(g)) = I_g$ by the same reasoning. Thus $\widetilde{I}$ and $\mathcal{I}|_U$ agree (as ideals) on a basis of opens and hence are equal. Therefore $\mathcal{I}$ is a quasi-coherent sheaf of ideals.

At this point we may define Z as the closed subspace associated to the sheaf of ideals $\mathcal{I}$. For every affine open $U = \mathrm{Spec}(R)$ of X we see that $Z \cap U = \mathrm{Spec}(R/I)$ where I is a radical ideal and hence Z is reduced (by Lemma 01J2 above). By construction the underlying closed subset of Z is T. Hence we have found a closed subscheme with properties (a) and (b).

Let $Z' \subset X$ be a second closed subscheme with properties (a) and (b). For every affine open $U = \mathrm{Spec}(R)$ of X we see that $Z' \cap U = \mathrm{Spec}(R/I')$ for some ideal $I' \subset R$. By Lemma 01J2 the ring R/I' is reduced and hence I' is radical. Since $V(I') = T \cap U = V(I)$ we deduced that $I = I'$ by Algebra, Lemma 00E0. Hence Z' and Z are defined by the same sheaf of ideals and hence are equal. $\square$